

Computer Networks (CN)

- Large number of **separate** but **interconnected** computers - *computer networks*.
- Two computers are said to be interconnected if they are able to exchange information.
- Autonomous and Heterogeneous.
- How systems are interconnected ?
 - Twisted Pair Cables, fiber optics, microwaves, infrared and communication through satellites.
- Networks come in many sizes, shapes and forms
 - Local Area Networks (LAN)
 - Metropolitan Area Networks (MAN)
 - Wide Area Networks (WAN)
 - Internet

History of CN

- Information Exchange in olden days
 - In person delivery of the message
 - Line of Shouting Men on Tall Structures, Rhythmic Drums, Trumpets
- ARPANET 1969 - First Wired Network - Used in Department of Defense
 - The first message in the ARPANET - “lo”
 - Login
 - Then it crashed. The team fixed the bug within an hour and resend the very first meaningful message “login”
- Norman, 1970s - ALOHAnet
 - Universities of Hawaiian islands
 - First Wireless Network

Aloha Communications System Connects Hawaii to Arpanet

By Sasha Yasinin
CW Hawaii Correspondent

HONOLULU — The University of Hawaii's Aloha system is scoring innovative advances in satellite computer communications with an experimental satellite system which went on-line here recently.

The university also is involved in a joint study with the Advanced Research Projects Agency (Arpa), Bolt Beranek and Newman, Xerox and UCLA to design

Comsat ground station in sight of the satellite.

Aloha's director, Norman Abramson, sees this use of voice-grade channels as a big money-saver for computer communications.

As an example of the system's cost-saving potential, Abramson pointed to the possibilities in the next generation domestic satellite system.

"Each of these satellites," he told *Computerworld*, "will have 12 transponders with a bandwidth of 45MHz or so. If one of the potential carriers were to devote one transponder on the satellite to packet communications, it would be possible to transmit data at about 10 million baud. This single transponder could handle some five million on-line terminals.

"Converting to this kind of communications could save a lot of money. For example, Arpa spends in excess of \$1 million per year for line charges. With this much money it could buy a transponder on a domestic satellite.

"If computer users are ever going to be free of incredibly high costs of transmitting data, some system similar to this has got to be set up," Abramson said.

But Abramson foresees difficulties with potential carriers in the next generation system devoting a transponder to packet data communications. He sees problems with regulatory agencies in determining how to charge for the service.

Experimenting With Aloha

suitable protocol for packet communications via satellite.

As part of this project, Aloha becomes the first subscriber to a digital communications subsystem installed between the Comsat ground stations at Paumotu, Hawaii, and Jamesburg, Calif., last year.

With this installation Aloha becomes the first operational satellite node on Arpanet, via the Intelsat IV satellite over the Pacific.

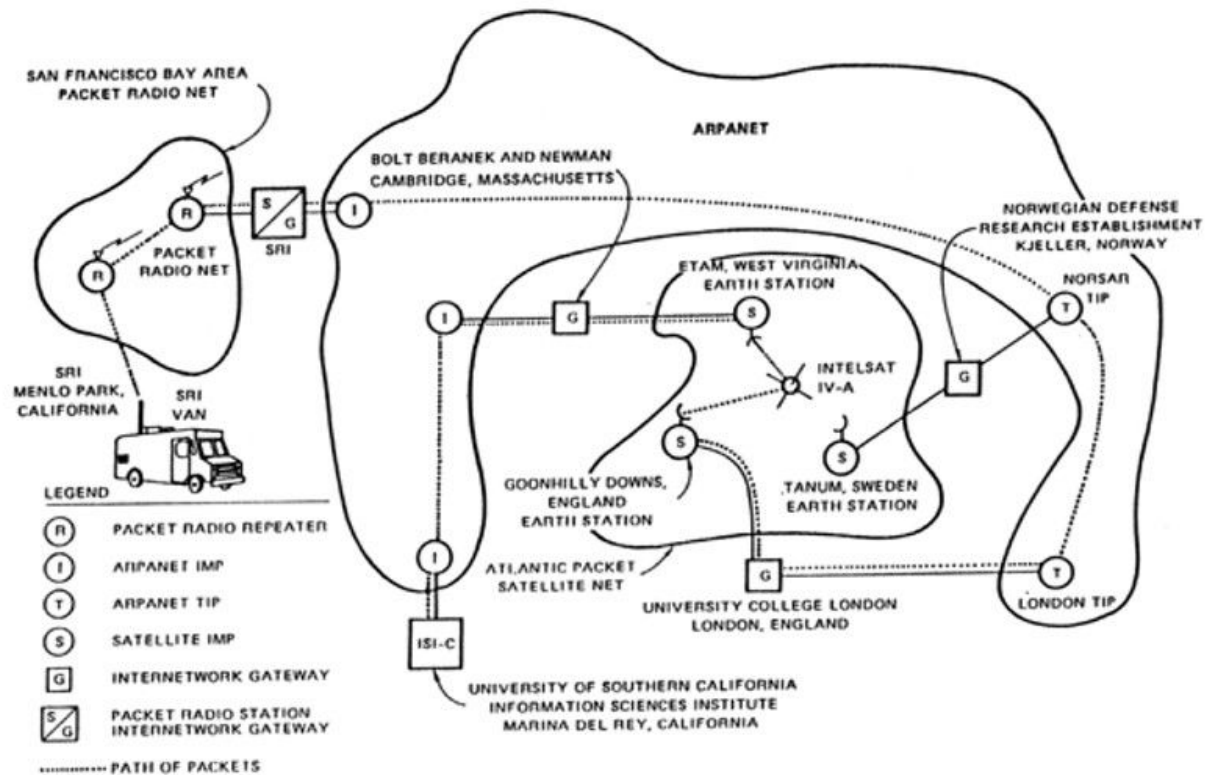
The Comsat system enables data to be transmitted at the rate of 50 kbit/sec over a single voice-grade channel.

The single satellite channel can provide two up-link and two down-link data channels and each of these four channels could be simultaneously available to any

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History of CN

- PRNET – Packet Radio Network (DARPA)
 - Wireless Network
- SATNET - Atlantic Packet Satellite Network - First Satellite Network
- SURAN – Survivable Radio Network



1977

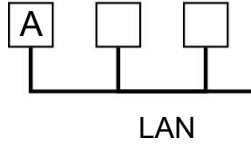
- IETF – Internet Engineering Task Force - 1986
- Birds-of-a-Feather (BoF) – IETF Discussion form for improving Quality of Service (QoS).
 - 2013 - QoS and QoE
- Different Parameters Defined:
 - Performance
 - Response Time, Fairness,
 - Reliability
 - Availability, Robustness
 - Security
 - Data and Hardware
- Different Generations of Computer Networks
 - 1G, 2G, 3G, 4G, and 5G and Beyond

Types of Network

Based on the Scale - Area covered

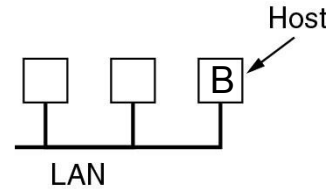
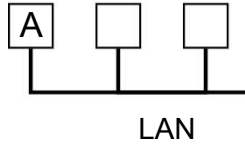
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	

Internet - A bird's eye view

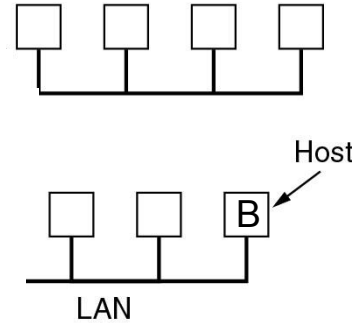
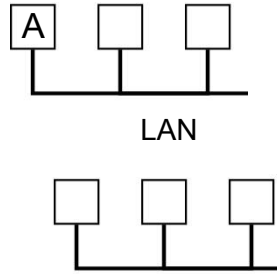


Host, Nodes, System, Computer, Server

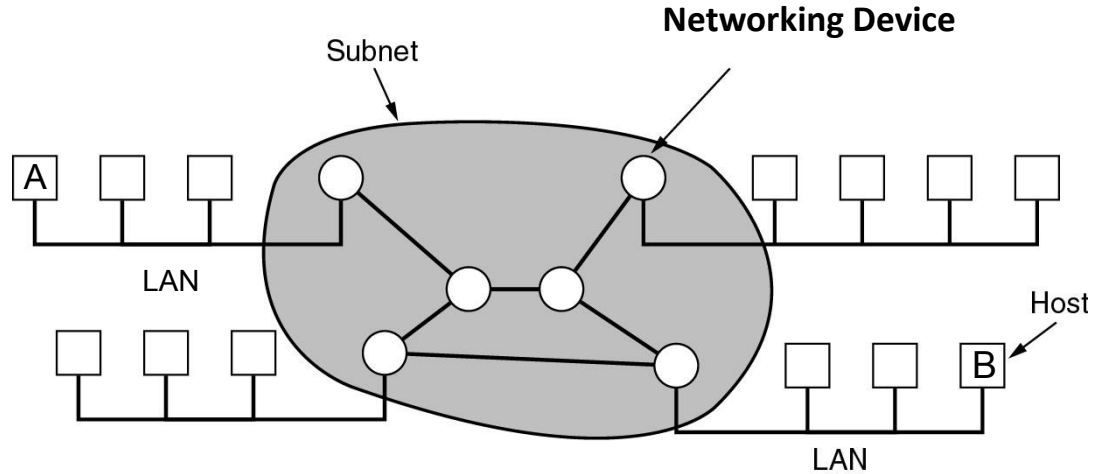
Internet - A bird's eye view

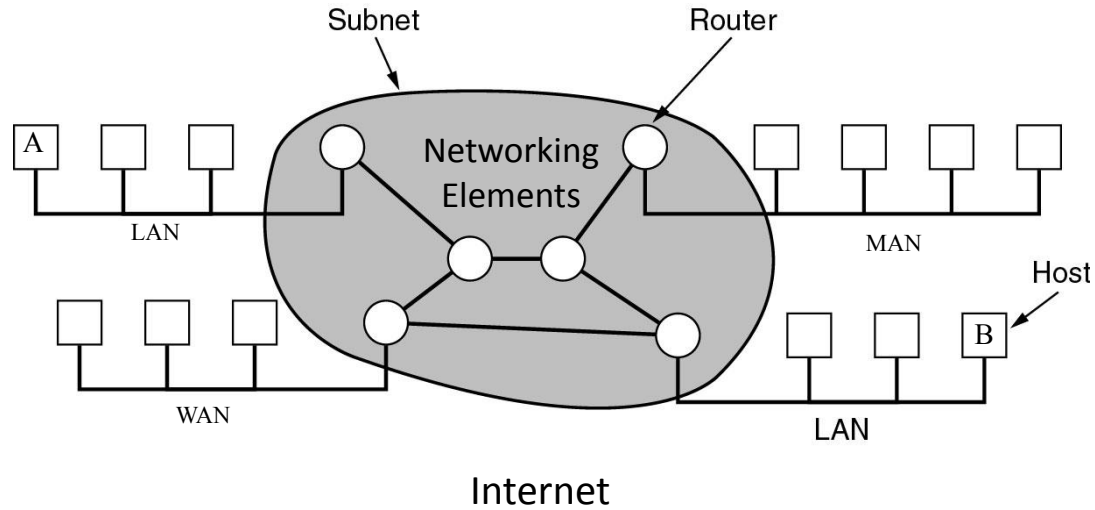


Internet - A bird's eye view



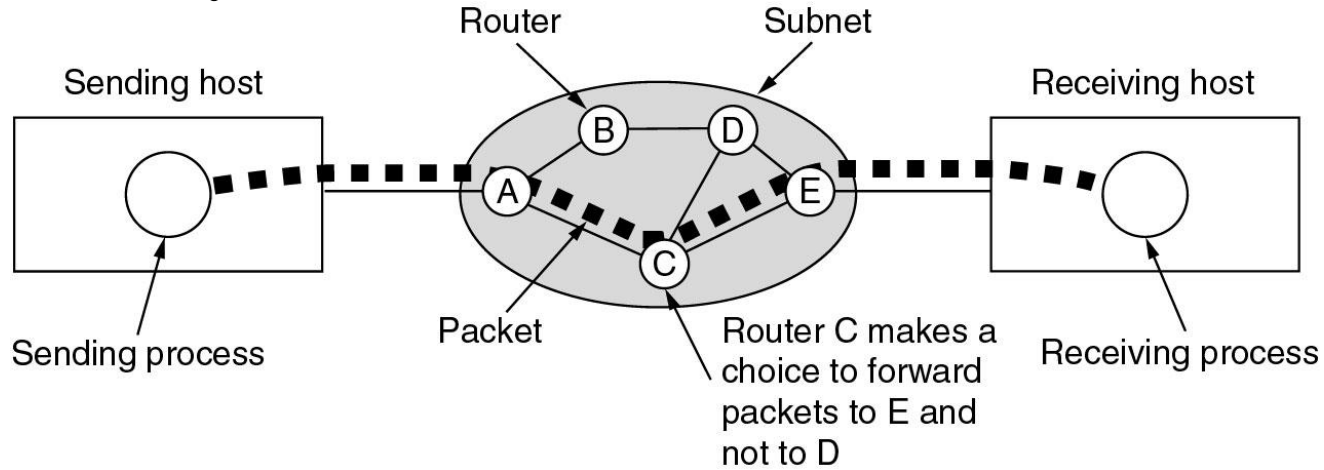
Internet - A bird's eye view





- What is Internet ?

Behind the System



- Often multiple routes/paths, of different lengths will be available.
- Finding good routes/paths is important.

Internet and Intranet

- Internet
 - Open to all.
 - Spread across the world
- Intranet
 - Restricted - Private
 - Organization or firm
- NITK Has Internet or Intranet ?
 - We have both
 - Internet for Outside World
 - Intranet for only NITK'ns

Types of Network Based on Medium

- Wired Network
- Wireless Network

Thank You.